IUPAC naming

✓ Changes save automatically and apply to your next problem. ☐ Don't show this again **Got it**

Settings

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

Molecules

Shared by every generator. Pick what your course covers once, and IUPAC naming, Synthesis and future generators all use it, unless you customize a generator separately.

Allowed structures and functional groups — Select all Clear

Generators only build molecules from what's checked.

Carbon skeleton

- ☑ Alkane
- ☑ Alkene
- ☑ Alkyne
- ☐ Aromatic

Functional groups

- ☑ Alcohol
- ☑ Alkyl halide
- ☑ Ether
- ☐ Thiol
- ☐ Aldehyde
- ☐ Ketone
- ☐ Epoxide

Carboxylic acid derivatives

- ☐ Carboxylic acid
- ☐ Acid chloride
- ☐ Anhydride
- ☐ Ester
- ☐ Amide
- ☐ Nitrile

Auxiliary

- ☐ Amine
- ☐ Nitro

Molecule size
Number of carbons in each molecule. 3 to 10

Include rings
Cycloalkanes and other cyclic molecules.

Include heteroatoms in rings
Pyridine, furan, thiophene and similar. Needs Aromatic checked.

Stereochemistry
When included, answers must give correct R/S and E/Z. Off | Include

IUPAC naming

Changes save automatically and apply to your next problem. Don’t show this again Got it

Settings

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

IUPAC naming

Settings that only apply to the IUPAC naming generator.

MOLECULE OPTIONS

Use the shared Molecules settings
Semester 1 · 7 groups · 3–10 carbons · rings on · stereo off · Edit shared

Customize for IUPAC naming only
Changes here won’t affect other generators.

NAMING PROBLEMS

Parent chain length
Ring size for aromatic parents.
3 to 10

Most senior functional group
Chosen from the groups allowed under Molecule options.
Any allowed

Number of principal groups
1 to 2

Number of secondary groups
0 to 2

PRACTICE

Count skipped problems in my score

Keyboard shortcuts
Enter submits · Enter again for next · Esc skips

IUPAC naming

✓ Changes save automatically and apply to your next problem. ☐ Don't show this again **Got it**

Settings

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming !

Synthesis

Acid/Base Coming soon

IUPAC naming

Settings that only apply to the IUPAC naming generator.

NAMING PROBLEMS

Parent chain length
Ring size for aromatic parents.
4 to 3

! Minimum is larger than maximum.

Most senior functional group
Chosen from the groups allowed under Molecule options.
Carboxylic acid

! Carboxylic acid isn't in your allowed groups. Allow it

Number of principal groups
0 to 2

Number of secondary groups
0 to 2

PRACTICE

Count skipped problems in my score

Keyboard shortcuts
Enter submits · Enter again for next · Esc skips

IUPAC naming

Settings

PRACTICING Alk

Problem 1

MOLECULE

IUPAC NAMING · QUICK START

Choose what to practice

We've picked what most courses cover first. Adjust it to match yours, or just start.

Carbon skeleton

Pick at least one

- [x] Alkane
- [x] Alkene
- [x] Alkyne
- [ ] Aromatic

Functional groups

Leave all unchecked for plain hydrocarbons

- [x] Alcohol
- [x] Alkyl halide
- [x] Ether
- [ ] Thiol
- [ ] Aldehyde
- [ ] Ketone
- [ ] Epoxide

> **More functional groups** Carboxylic acid derivatives, amine, nitro

> **Size, rings and stereochemistry** 3–10 carbons · rings on · stereochemistry off

> **Advanced options** 2°/3° counts, ring topology, aromatic positions, group count and distance, custom fragments

Saved to Settings › Molecules. Change it any time.

Start practicing →

Name this mole

Type the I

Skip

Next problem →

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

Molecules

Shared by every generator.

Advanced options

	CARBONS		AMINES (needs Amine checked)	
Secondary (2°) Carbon bonded to 2 carbons · R_2NH	0	to 4	0	to 1
Tertiary (3°) Carbon bonded to 3 carbons · R_3N	0	to 2	0	to 1

Alkyl substituents (R)

Select all Clear

Which alkyl substituents can branch off the parent chain or ring.

1 carbon

- ☑ **Methyl** — methyl

2 carbons

- ☑ **Ethyl** — ethyl

3 carbons

- ☑ **Propyl** — propyl
- ☑ **Isopropyl** — propan-2-yl

4 carbons

- ☑ **Butyl** — butyl
- ☑ **Isobutyl** — 2-methylpropyl
- ☑ **sec-Butyl** — butan-2-yl
- ☑ **tert-Butyl** — 2-methylpropan-2-yl

5 carbons

- ☑ **Pentyl** — pentyl
- ☑ **Isopentyl** — 3-methylbutyl
- ☐ **Neopentyl** — 2,2-dimethylpropyl
- ☑ **tert-Pentyl** — 2-methylbutan-2-yl
- ☐ **sec-Pentyl** — pentan-2-yl
- ☐ **3-Pentyl** — pentan-3-yl

☑ Allow more complicated alkyl groups

Longer or more highly branched side chains than the ones above, such as 2-ethylbutyl or 3,3-dimethylbutyl.

Functional groups in these side chains

- ◉ **Not allowed** — Side chains are carbon and hydrogen only.
- ○ **Allowed** — Side chains can carry the functional groups checked in Molecules, e.g. 2-hydroxyethyl or 3-chloropropyl.

Ring topology

Single rings are always included when Include rings is on.

☐ Spiro ☐ Fused ☐ Bridged

Aromatic ring positions

Substituent patterns on disubstituted benzene rings. Needs Aromatic checked.

☐ Ortho (1,2) ☐ Meta (1,3) ☐ Para (1,4)

Functional group count

Total number of functional groups in each molecule.

0 to 3

Functional group distance

How far apart two functional groups are (Warren, *The Disconnection Approach*).

☑ 1,1 ☑ 1,2 ☑ 1,3 ☑ 1,4 ☑ 1,5

Custom fragments

Every generated molecule must contain all of these fragments.

+ Add fragment

C1CCCCC1

C(=O)OC

Edit Remove

IUPAC naming

✓ Changes save automatically and apply to your next problem. Don't show this again Got it

Settings

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

Molecules

Shared by every generator.

eeds Amine checked)

to 1

to 1

Fused Bridged

Meta (1,3) Para (1,4)

0 to 3

1,3 1,4 1,5

+ Add fragment

t checked above.

Allow ester

Add a custom fragment

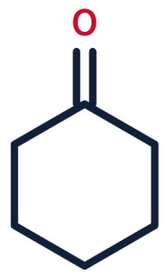
Draw | Enter SMILES

SMILES

O=C1CCCCC1

One fragment per entry, e.g. C1CCCCC1 for a cyclohexane ring.

PREVIEW



Cyclohexanone fragment · Open in Draw to edit

Cancel Add fragment

ACCOUNT

Sign in

Settings

ALL GENERATORS

Molecules >

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Synthesis Laptop or tablet >

Acid/Base Coming soon

Changes save automatically and apply to your next proble

Don't show this again

Got it

Use shared Molecules

Semester 1 · 7 groups · Edit shared

Customize for IUPAC only

NAMING PROBLEMS

Parent chain length 3 to 10

Most senior group

Any allowed

Edit

Principal groups 1 to 2

Secondary groups 0 to 2

PRACTICE

Count skipped problems